**Project Idea Proposal**

Object Oriented Programming (Java)


**Team members:**

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**Course Instructor:**

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**Project: Banking Application with OOP in Java**

1. Introduction:

   We are making a computer program  for banking . It will have buttons and windows that make it easy for people to use. The goal is to help people manage their money better and do banking stuff like sending money, checking balance, paying Bills, mobile loads.

2. Problem Statement

   Classic Banking Application will be developed often lack modern features and are not built using efficient programming paradigm.

   This project aims to address these shortcomings by creating a Banking Application using Java and OOP principles. The goal is to develop  a user-friendly visually appealing app with enhanced functionalities such as dynamic backgrounds, effects receiving and transferring money and finance managing system.

## 3.Proposed Methodology

- ***Easy to use:*** Make it simple for sending for people to understand and  use.
- ***Do Important Things :*** let people do basic banking tasks like checking their balance, sending money, paying Bills, mobile loads.
- ***Keep Things safe:*** Make sure people's personal info and money are kept safe.

- ***Get better overtime:*** Make the program in a way that we can add more features later if needed.
- ***Work Well Always:*** Make sure it doesn't break or have problems when people use it.
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## *4.Objectives:*

- ***Easy to use:*** Make it simple for sending for people to understand and  use.
- ***Do Important Things :*** let people do basic banking tasks like checking their balance, sending money, paying Bills, mobile loads.
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## 5.Technologies

- ***Java:*** We're using a programming language called java because it's good for making programs that work on different type of computers.
- ***Swing GUI:***  We are using a tool called Swing to make the buttons and windows look nice and easy to use.
- ***Database:*** We'll use a special system to store and manage people's banking info securely.

- ***Security:*** We'll add things to keep people's info safe, like making sure only the right people can access the program.
- ***Networking:*** We'll make it so the program can talk to the bank's computers to do things like check balances and send money.

### *Architecture:*

- ***Model-View-Controller (MVC):*** We'll organize the program into three parts: one for handling data and logic, one for showing things on the screen, and one for controlling how people interact with the program. This helps us keep things organized and easy to work on.
- ***Layered Architecture:*** We'll split the program into different parts, like one for showing things, one for doing the banking stuff, and one for saving and getting data. This makes it easier to add new features later.
- ***Client-Server Model:*** We'll set it up so people's computers (the clients) talk to the bank's computers (the server) to do banking stuff. This way, everything stays organized and safe on the bank's side.

## Project Timeline

**Week 1-2**

Design and Implement core classes(Deposit, transfer, pay bills)

Integration of  basic banking Mechanism

Initial GUI setup and testing.

**Week 3**

Implement of Dynamic feature.

Refinement of user interface element

Testing and debugging

**Week 4**

Final testing and bug fixing

Documentation preparation

Submission of completed Project

## Conclusion

In conclusion, our project aims to develop a robust and user-friendly desktop banking application that addresses the shortcomings of existing solutions. By following structured methodology and leveraging appropriate technologies, we aim to provide users with a secure and efficient platform for managing their finances. Through diligent implementation and testing, we strive to deliver a high-quality product that meets the expectation of our users and stakeholders.

- **Project Proposal Video Link**
  https://drive.google.com/file/d/1D-v7GMo4OKX3yrnBxSusDa0GYn5LGXPb/view?usp=drivesdk